



Lab 7: Data Link Layer (MAC, CSMA and VLANs)

In this lab you will study three concepts related to the Data Link Layer: how switches learn MAC addresses and use them to forward traffic, how CSMA/CD manages collisions on shared networks, and how VLANs segment a network into isolated groups.

MAC Address Table and Switching

Every network interface card (NIC) has a MAC (Media Access Control) address, which is a unique 48 bit identifier written as six pairs of hexadecimal digits, for example 00:1A:2B:3C:4D:5E. MAC addresses are assigned by the device manufacturer and operate at Layer 2, meaning they are used to deliver frames between devices on the same local network. They are different from IP addresses, which are used to route traffic between different networks.

How a Switch Learns MAC Addresses

When a switch starts up, its MAC address table is empty. It fills this table automatically by watching traffic:

- When a frame arrives on a port, the switch reads the source MAC address from the frame and records it alongside the port number in its MAC address table.
- The next time a frame arrives destined for that MAC address, the switch forwards it only to the correct port instead of sending it out everywhere.
- If the switch has not yet seen a destination MAC address, it sends the frame out of all ports except the one it came in on. This is called flooding. Once the destination device replies, its MAC is learned and future traffic is forwarded directly.
- Each entry in the table has an ageing timer (300 seconds by default). If no traffic is seen from a device within that time, its entry is removed.

A Switch VS A Hub

A hub is a simple device that repeats every signal it receives out of all its ports. Every device connected to a hub sees every frame, regardless of who it was meant for, and all devices share the same bandwidth. A switch is smarter: it reads MAC addresses and forwards each frame only to the correct port. This means two pairs of devices can communicate simultaneously without interfering with each other.

Carrier Sense Multiple Access/ Collision Detection (CSMA/CD)

In early Ethernet networks, all devices shared a single cable. Only one device could transmit at a time, if two devices transmitted simultaneously, their signals would overlap and corrupt each other. This is called a collision. CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the set of rules that governs how devices take turns on a shared medium and recover when collisions do happen.



Although modern switched networks have largely eliminated collisions (each cable between a PC and a switch port is a private link), CSMA/CD still applies to any half-duplex (data can flow in both directions, but only in one direction at a time) link or any network segment that uses a hub.

How CSMA/CD Works

The Carrier Sense portion:

Before sending a packet, a device listens to the cable to check whether another device is already transmitting. If the cable is busy, it waits.

The Multiple Access portion:

All devices have equal access. There is no controller that decides whose turn it is.

The Collision Detection portion:

While transmitting, a device also listens to the cable. If the signal on the cable differs from what it is sending, it knows a collision has occurred.

When a collision is detected:

- Both devices immediately stop transmitting and send a short jam signal to make sure all devices on the network know a collision occurred.
- Each device waits for a random amount of time before trying again. The randomness means they are unlikely to collide a second time.
- After the wait, each device listens again and retransmits when the cable is clear.

Collision Domains

A collision domain is the group of devices that share the same physical medium and can interfere with each other. The table below summarises how different network devices affect collision and broadcast domains:

Device	Collision Domain	Broadcast Domain
Hub	Single (all ports share one)	Single (all ports share one)
Switch	Per port (isolated per link)	Single (all ports share one)
Router	Per port (isolated per link)	Per port (isolated per link)

Every port on a switch is its own isolated collision domain, which is why switches have replaced hubs in modern networks. With a switch, two devices can transmit at exactly the same time without any collision because the switch handles each link independently.



VLANs and Inter-VLAN Routing

A VLAN (Virtual Local Area Network) is a way of splitting a single physical switch into multiple isolated groups. Devices in the same VLAN can communicate freely with each other, but they cannot send traffic to devices in a different VLAN, it is as if they are on completely separate switches, even though they share the same hardware.

Normally, a router creates boundaries between separate networks. VLANs let a switch create those same boundaries in software, without needing extra hardware. Each VLAN is its own broadcast domain, meaning a broadcast sent by a device in VLAN 10, for example, will only reach other devices in VLAN 10.

VLANs are used for:

Security:

Sensitive departments (e.g., HR, Finance) can be kept separate from the rest of the network even if they share the same physical switches.

Reduced broadcast traffic:

Smaller groups mean fewer broadcast frames, which keeps the network fast.

Easier management:

Devices can be grouped by function rather than physical location.

Access Ports and Trunk Ports

There are two types of ports you'll need when configuring VLANs.

Access port:

Connects an end device (PC, printer, etc) to the switch. The port belongs to exactly one VLAN. The device itself does not need to know anything about VLANs.

Trunk port:

Connects two switches together, or a switch to a router. A trunk port carries traffic for multiple VLANs at once. Frames on a trunk link are labelled (tagged) with the VLAN they belong to so the receiving device can tell them apart.

802.1Q Tagging

The 802.1Q standard is how Cisco switches tag VLAN traffic on a trunk. A small label containing the VLAN ID is added to each frame. The receiving device reads this label, strips it, and forwards the frame to the correct VLAN.

Inter-VLAN Routing

Because VLANs are separate broadcast domains, they are also on separate IP subnets. A device in VLAN 10 (e.g., 192.168.10.x) cannot communicate with a device in VLAN 20 (e.g., 192.168.20.x) without going through a router.



Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: MAC Address Table and Switching (5 Marks)

The following are the steps and IP addresses you need to follow to observe MAC addresses on CISCO.

PC0 192.168.1.1 255.255.255.0

PC1 192.168.1.2 255.255.255.0

Step 1: Build the Topology

- Open Cisco Packet Tracer
- Add a 2960 switch and two PCs. Label them clearly.
- Connect PC0 to fa0/1 and PC1 to fa0/2 using Copper Straight-Through cables. Wait for the link lights to turn green.

Step 2: Assign IP Addresses

- On each PC, open Desktop > IP Configuration, select Static, and enter the address from the details above.

Step 3: Check the MAC Table Before Traffic

- Click the switch, open the CLI tab, and press Enter to get a prompt.
- Enter privileged EXEC mode and run

```
Switch> enable  
  
Switch# show mac address-table
```

- The table should be empty.

Step 4: Send a Ping

- Open a Command Prompt on PC0 (Desktop > Command Prompt) and run:



```
PC> ping 192.168.1.2
```

- The first ping may time out while the ARP exchange takes place. Subsequent pings should succeed.

Step 5: Check the MAC Table After Traffic

- Return to the switch CLI and run *show mac address-table* again.
- Both MAC addresses should now appear, each mapped to its port.

The MAC address table goes from empty to populated after the ping. The switch learned PC0's MAC when PC0 sent its ARP request, and PC1's MAC when PC1 replied. From this point, frames between PC0 and PC1 are forwarded directly to the correct port rather than flooded to all ports.

Task 2: CSMA/CD Simulation (5 Marks)

The following are the steps and IP addresses you need to follow to observe CSMA/CD on CISCO.

PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Step 1: Build the Topology

- Add Hub0 (Hub-PT) and four PCs. Connect each PC to the hub using Copper Straight-Through cables.
- Assign the IP addresses from the details above to each PC.

Step 2: Switch to Simulation Mode

- Click the Simulation button.
- In the Event List Filters, make sure ICMP is selected.

Step 3: Send Simultaneous Pings

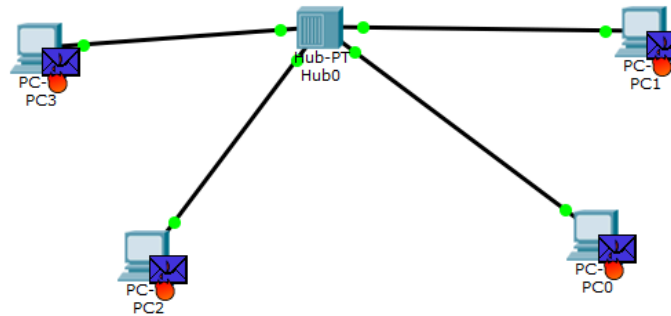
- From PC0, initiate a ping to PC1.
- While the ping is in progress, from PC2, initiate a ping to PC3. **(use the simple PDU tool)**



The goal is to have both pings in flight at the same time so their frames collide on the shared hub.

Step 4: Observe the Collision

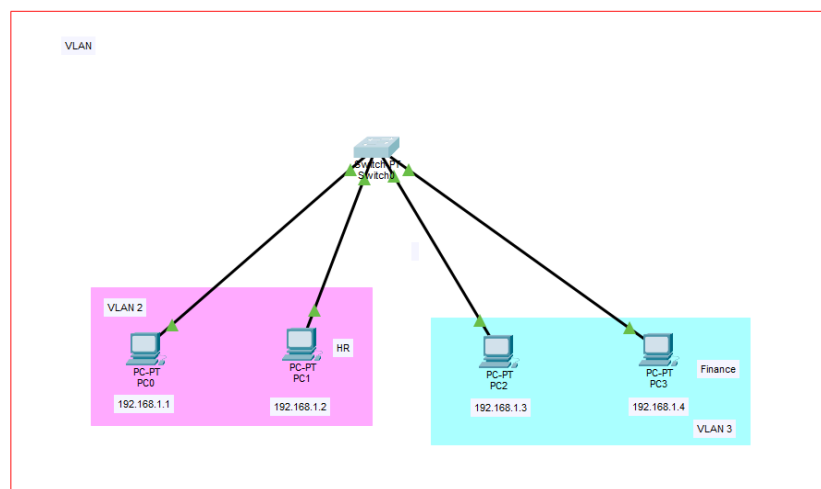
- In the Simulation Panel, observe the events. Collisions should occur due to the hub's broadcasting nature, illustrating CSMA/CD in action.



Because all four PCs are in the same collision domain (connected via the hub), frames from PC0 and PC2 collide when sent at the same time. The jam signal propagates to all ports, both senders wait a random time, and one of them retransmits successfully. Notice that PC1 and PC3 also receive the jam signal even though they were not sending, this is the nature of a shared medium.

Question: How is collision handled in CSMA/CD?

Task 3: Implementing VLAN (5 Marks)





The following are the steps and IP addresses you need to follow to observe VLANs on CISCO.

PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Step 1: Build the Topology

- Add Switch0 (Switch-PT) and four PCs. Connect each PC to the switch using Copper Straight-Through cables. **(ensure you assign the interfaces as they appear in step 2)**
- Assign the IP addresses from the details above to each PC.

Step 2: Create VLANs on the Switch and assign them to the PCs

- Open **Switch0** and go to the **CLI** tab. Run the following commands.

```
Switch>enable
Switch#config t
Switch(config)#vlan 2
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet1/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet2/1
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#interface fastEthernet3/1
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#enable
Switch#show vlan brief
VLAN Name Status Ports
-----
1 default active Fa4/1, Fa5/1
2 HR active Fa0/1, Fa1/1
3 Finance active Fa2/1, Fa3/1
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
```



Step 3: Observe VLAN Isolation

- From PC0, initiate a ping to PC1 (192.168.1.2). The ping will be successful because both PCs are in VLAN 2.
- From PC0, initiate a ping to PC2 (192.168.1.3) or PC3 (192.168.1.4). The pings will fail because these PCs belong to VLAN 3.

Submission Guidelines (5 Marks):

- Submit a single Word/PDF document containing all tasks and the CISCO packet file(s) to the GCR, **also ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1**.
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1**. **(for multiple files add the task number at the end)**
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

